

HS 312 – Introduction to Science and Technology Studies

Lecture 14
Mahendra Shahare

Interests and rhetoric

- Controversy – claims about the stakes, strategies, weaknesses, and resources of opponents
- Interest models – Why protagonists take unorthodox position? Recognition, monetary gain, passion for challenging orthodoxy?
- What tools do actors employ to further their positions?
- **Rhetorical (persuasive) tools** are central to convince people of claims; usually a published paper
- Empirical studies are designed with persuasion in mind

Interests and rhetoric

- Rhetoric – Convince a particular audience of some fact(s), legitimacy of one's positions by making one's own work appear more scientific
- The idea of Science itself is most important rhetorical resource i.e. method, data, empirical, falsifiable etc.
- Disciplines are important too e.g. cold fusion case (chemistry vs physics)
- Reputation – brilliant theorist/experimenter, work experience with a respected colleague, affiliation with a large research institution, PI of a large laboratory
- Invocation of norms to delegitimise opponents e.g. financial link and disinterestedness
- Scientific paper – may contain criticisms of assumptions, studies, experiments, or arguments made by opponents in the controversy

Resolutions of Controversies

- How are disputes in S&T resolved?
- Critique of observations, experiments, and positions e.g. consistency and plausibility
- New tests and calibration of instruments
- Isolating one position as more scientific or central e.g. solidify agreement amongst core-set of researchers whose opinion counts most
- Showing one position to be more central e.g. ideas become dominant because many researchers can see how to use it, how to build on it, regardless of its validity
- Ignoring deviant viewpoints and data e.g. If position contradicts central beliefs

Summary

- While work in **STS is sometimes viewed as attacks on legitimacy of S&T**, but the point of STS studies is to understand sources and meanings of knowledge
- Symmetrical approach is intended to show that **disagreements can be legitimate** and heterodoxy is tenable
- Evidence is tied to local culture and contexts – data is given meaning as evidence by the people who make use of it
- But controversy studies also show **the unruly process of arguing**
- Controversy studies are viewed as supporting unorthodox position
- **The politics of STS - Risk of being ‘captured’** / appropriated by participants e.g. Richards (1996) reports that her work on Vitamin C and cancer is viewed as supporting alternative medicine

Group Presentations

7	22B1023	Chandrakant Pradhan	Chapter 7	17	210020111	Saatwik Amrawat			
			Set the controls for the heart of the sun: the strange story of the missing solar neutrinos	21-Feb					
	22B1059	Darisipudi Saranya			23B2135	Sabarish S			
	22B1063	Dayyala John Joseph			22B1213	Sachi Mukesh Deshmukh			
	22B3904	Devtanu Barman			210020114	Salil Singhal			
	210110000	R Sabita			22B0515	Prasanna Omesh Karthikeya			
10	22B1276	Kaushal Deepak Malpure	Additional Reading Bern	20	22B2197	Vaishnavi Arun Kukde			
			Lady Lovelace's Objection: The Turing–Hartree Disputes	21-Feb					
	22B0952	Kukudala Sai Aditya			22B1040	Veebhuti Karthikeya			
	22B1060	Kunal Chaudhari			22B0306	Vibhor Sharma			
	22B0622	Lalit Meena			210260059	Vishal Kumar			

Sources

- Sismondo Sergio (2010). Questioning Functionalism in the Sociology of Science. Chapter 3, in *An Introduction to Science and Technology Studies*; second edition: Blackwell Publishing, Oxford UK.
- Harry Collins and Trevor Pinch (1998). Set the controls for the heart of the sun: the strange story of the missing solar neutrinos. Chapter 7, in *The Golem: what you should know about science*; second edition: Cambridge University Press, Cambridge UK.
- Bernardo Gonçalves (2024). *Lady Lovelace's Objection: The Turing–Hartree Disputes Over the Meaning of Digital Computers, 1946–1951*. IEEE Annals of the History of Computing, Jan.-Mar. 2024, vol. 46, pp. 6-18.

Thank you